Warming and survey effort contribute comparably to new bird distribution records in China

— once each record is dated to its discovery

1,029
provincial new records
CBNR dataset, 2000–2025

175,901
species × province × year
cells in the risk set

649
first records modelled
392 species · 31 provinces

Chenchen Ding

Institute of Ecology, College of Urban and Environmental Sciences, Peking University
International Institute for Applied Systems Analysis (IIASA)

Third Asian Ornithological Conference · National University of Singapore · 14–17 November 2026

A new record can mean the bird moved — or that someone finally looked

Every record needs a bird and a birder

Range maps are built from records.

A record = where birds are × where people look.

In the raw data, the two cannot be told apart.

In China, both grew fast after 2000

The climate warmed quickly.

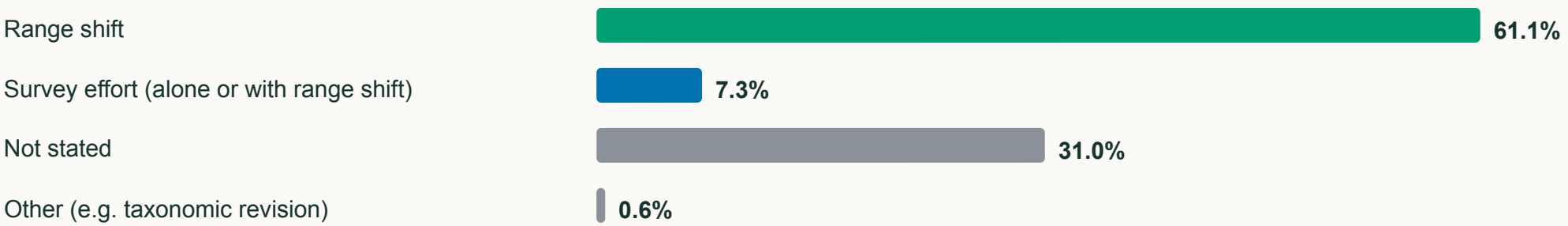
Birdwatching grew from a niche into a mass pastime.

A model with only one of them gives it the credit for both.

Our question

Are new records driven by real range change or by observation — and can we measure both in one model?

How the original papers explain the 1,029 CBNR records



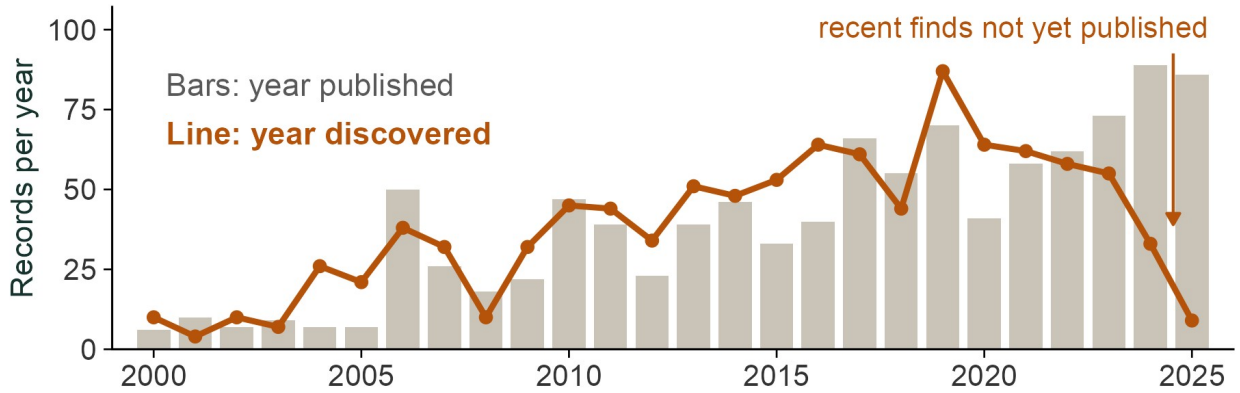
Source: CBNR v2, field ‘Inferred discovery reason’

CBNR: an open dataset of provincial new bird records in China, 2000–2025

a 1,029 records across 33 provincial units



b Records reach print 2.1 years after discovery



What is in it

1,029 records · 568 species · 23 orders

33 provincial units · 2000–2025

Mined from peer-reviewed papers by an LLM-assisted pipeline; every record then checked by hand

Why it matters for this talk

Discovery date and publication year

Georeferenced; three taxonomies linked

Detection method, discovery reason, status

Open data · CC BY 4.0

Zenodo, v2 (June 2026)

doi.org/10.5281/zenodo.20809949

Scan to download →

One hazard model, four parts that cannot be dropped

For every species, in every province it could reach, in every year: is this the year of its first record?

first record ~ warming × effort + climate variability + offset(log completeness)
+ (1 | species) + (1 | province) + (1 | province × year)

binomial, cloglog link

cloglog = discrete-time proportional hazards, so exp(β) is a hazard ratio (HR)

Complete risk set

Only provinces the species could reach
Leaves the risk set after its first record
Denominator = real opportunities

Dated to discovery

Publication lags discovery by 2.09 years
83% of events move to another year
An offset corrects recent incompleteness

Species-referenced warming

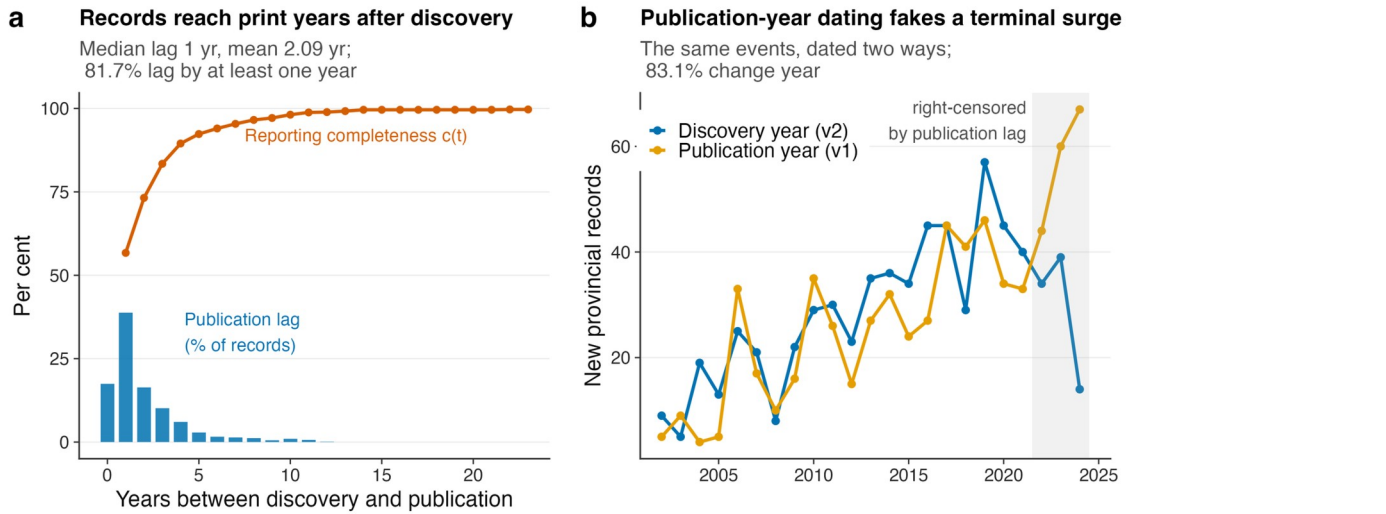
Province anomaly − range anomaly
15-year moving window
1 SD = 0.179 °C

Province × year effect

The scale at which surveys happen
Bird races, survey campaigns, new reserves
The observation process, modelled

Hazard ratio (HR): how many times one SD of a driver multiplies the yearly chance of a first record

Finding 1 · Change the dating, and the conclusion changes



Publication year → discovery year

Effort HR 1.79 → 1.31 (inflated by 37%)

Climate variability 0.85 → 0.97 (gone)

Accumulated warming 1.39 → 1.44 (stable)

The 0.85 had P = 0.00018 — a phantom

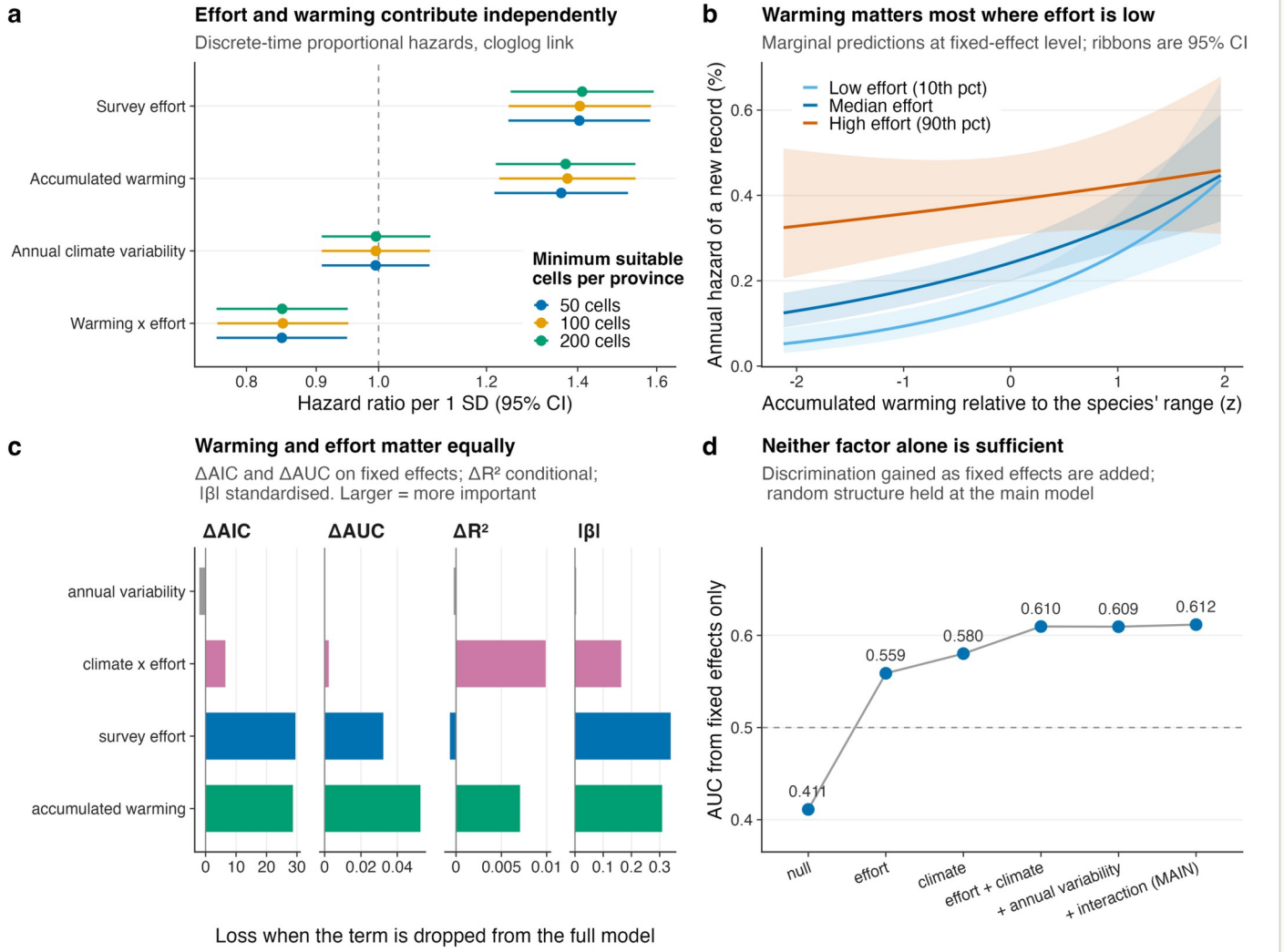
Why it happens

The lag is right-skewed (mean 2.1 years)
and grew along with birding databases,
so publication year = a year of more effort

So what?

Dating is a modelling decision,
not bookkeeping

Finding 2 · Warming and effort matter equally — and interact



Main model (n = 175,901)

Warming HR 1.36 (1.22–1.52)

Effort HR 1.40 (1.25–1.58)

Warming × effort HR 0.85 (0.76–0.95)

Climate variability HR 1.00 (no effect)

Read each effect against the right null

Permutation null for warming = 1.16, not 1

→ species-specific increment ≈ ×1.18

Shuffle effort's timing: it falls to 0.98

→ its signal comes from real timing

Province-clustered bootstrap

Warming 1.20–1.72 · effort 1.21–1.76

Both lower bounds above 1

The balance holds across four criteria and twelve specifications

Cost of dropping each term

ΔAIC: warming 28.7 · effort 29.5

ΔAUC: warming 0.053 · effort 0.032

Δ conditional R^2 and |β| agree

Discrimination ladder (Figure 1d)

Null 0.41 → effort only 0.56

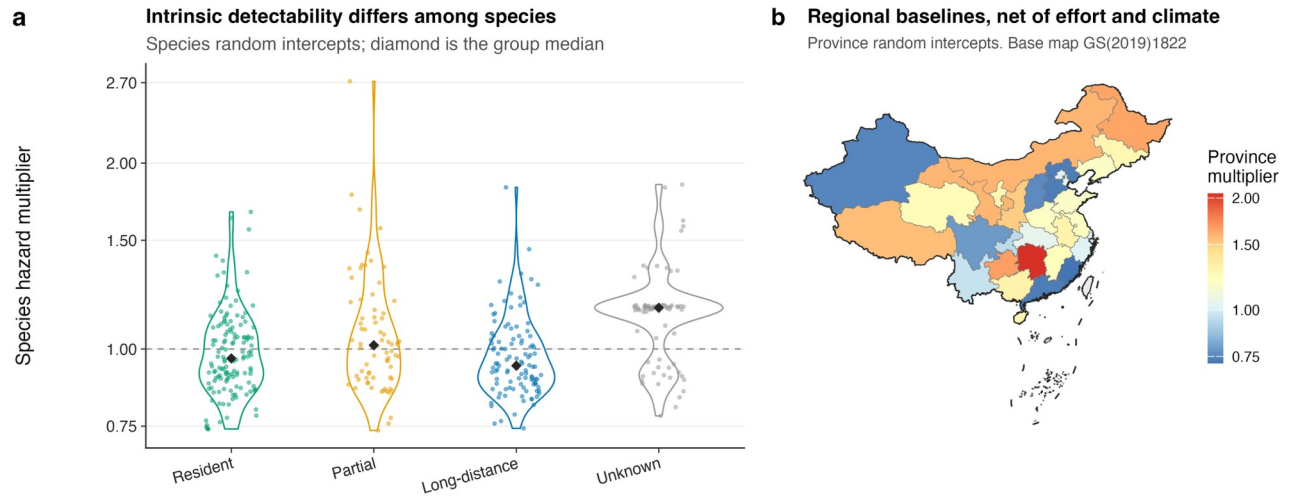
→ warming only 0.58 → both 0.61

Climate alone beats effort alone; both needed

Twelve specifications, one answer	Survey effort	Accumulated warming
SDM thresholds (50 / 100 / 200 cells)	**HR 1.40–1.41**	**HR 1.36–1.38**
4 effort proxies × 3 missing-data rules	**HR 1.25–1.52**	**HR 1.35–1.37**
4 climate windows × 3 baselines	—	**best window 18–20 yr**

A weather-driven process predicts the opposite window pattern — we did not see it.

Finding 3 · Survey campaigns move the record more than either driver



Province × year effect

× 2.23 per 1 SD

Three layers

Species: ×1.50 (detectability)

Province: ×1.39 (regional baseline)

Province × year: ×2.23 (survey shocks)

Adding this layer

Conditional AUC 0.72 → 0.85

Conditional R^2 0.27 → 0.41

Fixed effects unchanged

Treat it as noise,

and survey history reads as range change

Finding 4 · Effort is saturating; warming shows no such trend

Effort breaks proportional hazards (P = 0.00077): a finding, not a failure

Returns on effort are falling

2002: HR 2.18 (1.63–2.90)

2024: HR 1.07 (0.88–1.30)

After 2022 the interval includes 1

The easy gaps have been filled

Warming shows no such trend

Warming × time: P = 0.82

Time-varying model: warming HR 1.37

Effort mines a finite stock of gaps;
warming keeps adding new candidates

Consistent with a stock and a flow

The two time trends were not tested against each other: consistent, not proven

Same signal at province level

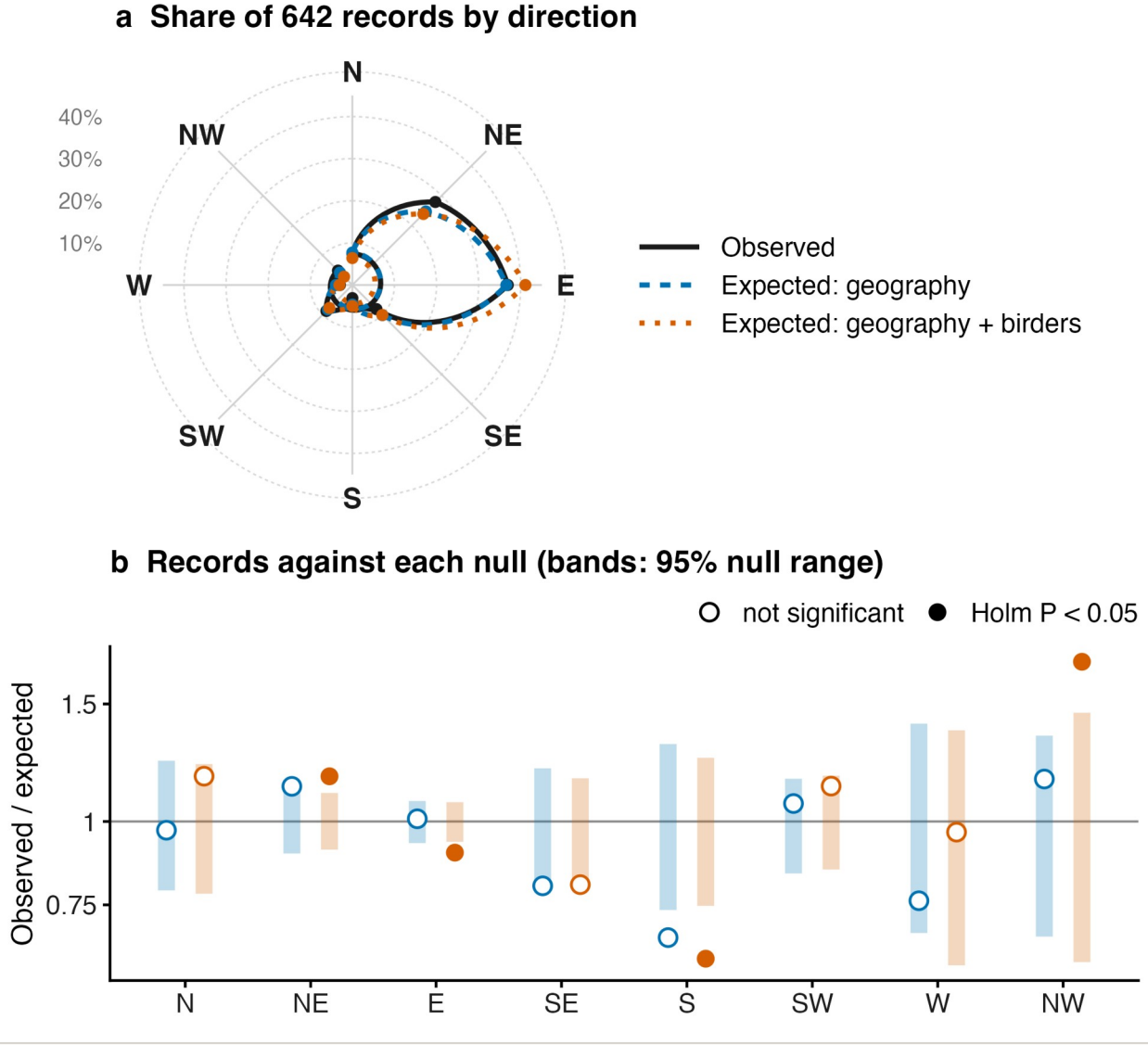
Provinces well covered in 2002–2008
now yield fewer records per unit of effort

IRR 0.71, P = 0.028

What it means for monitoring

More effort in eastern hotspots adds little;
send it to under-surveyed,
fast-warming regions

Finding 5 · The eastward bias is geography; records lean slightly north



Raw bearings (642 records)

East 36.9% · north-east 27.9%

$\chi^2 = 580$ against a uniform null,

but China lies east of most ranges

Null 1 · geography only

Each record placed at random among its species' possible provinces

East: 235 expected, 237 observed

Null 2 · geography + birders

More NE and NW, fewer E and S

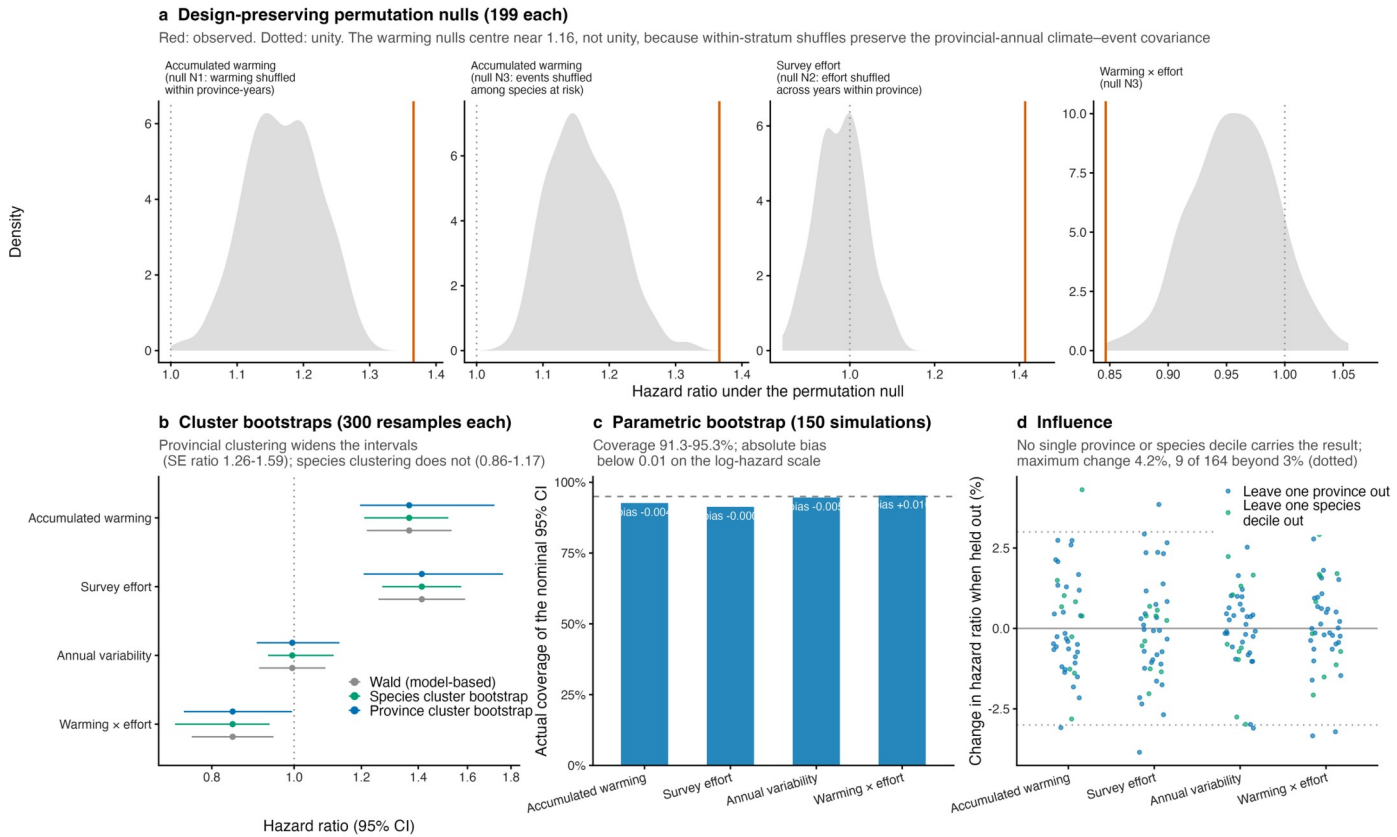
Northern sectors: 40% vs 33% expected

$\chi^2 = 25.8$, P = 0.001

Direction mostly reflects geography;

net of it, records lean slightly north

The results survive null models and resampling



Design-preserving nulls

Each null breaks one link, keeps the rest

199 permutations per null

All four coefficients outside their nulls

P = 0.005 (variability excepted)

Bootstraps and influence

Province-cluster bootstrap: SE ×1.26–1.59

Parametric bootstrap coverage 91–95%

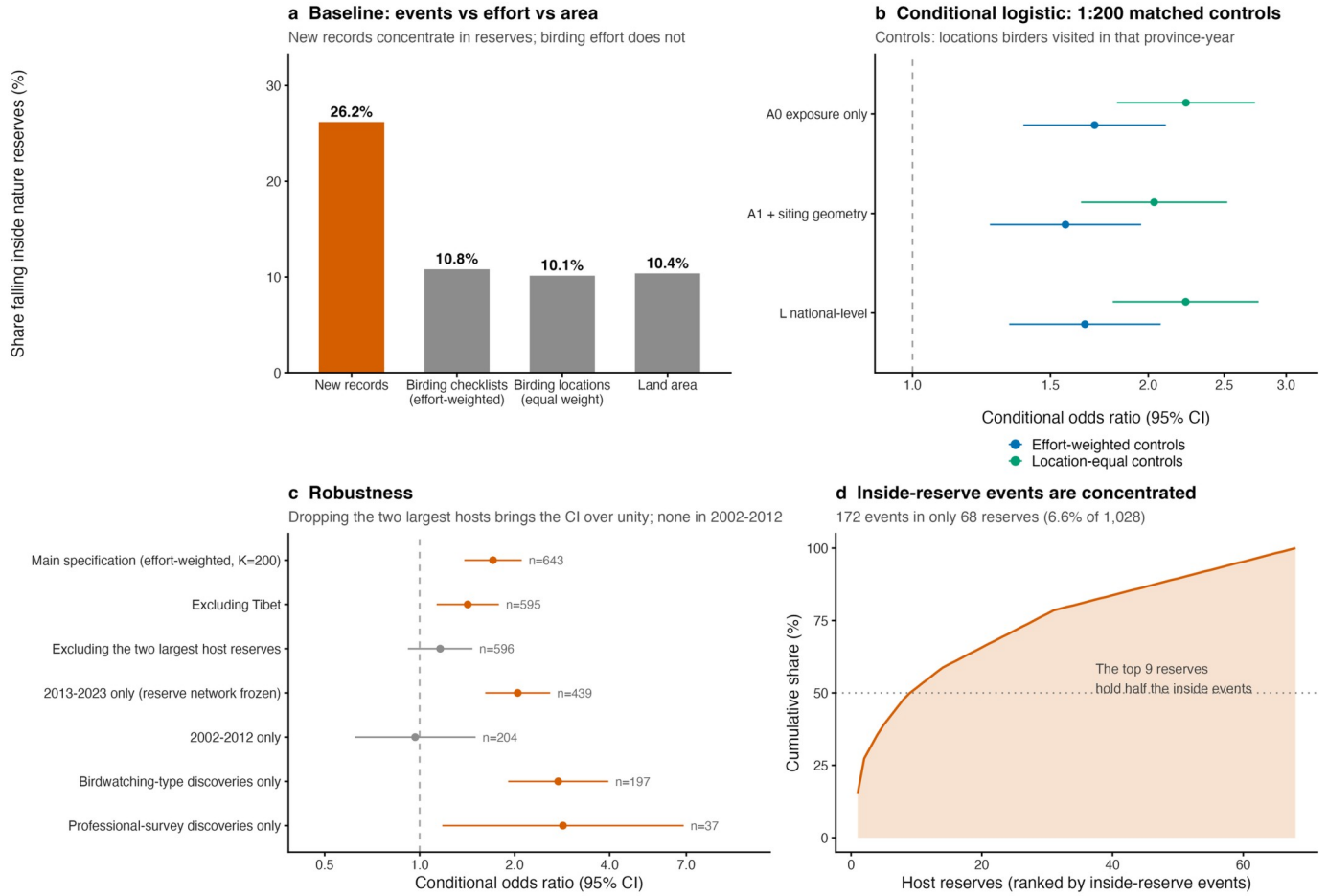
Leave one out: largest change 4.2%

No single province or species

group drives the result

Diagnostics (DHARMa): uniformity P = 0.74 · dispersion P = 0.59 · zero-inflation P = 0.59 · Moran's I P = 0.86

Use 1 · Reserves look rich in new records because birders go there



An apparent 2.5-fold enrichment

Records inside reserves: 26.2%

Birding effort inside reserves: 10.8%

Land area inside reserves: 10.4%

Against where birders actually went

Matched same province and year: OR 1.57

Without the 2 largest reserves: 1.16

Before 2013: no effect (0.97)

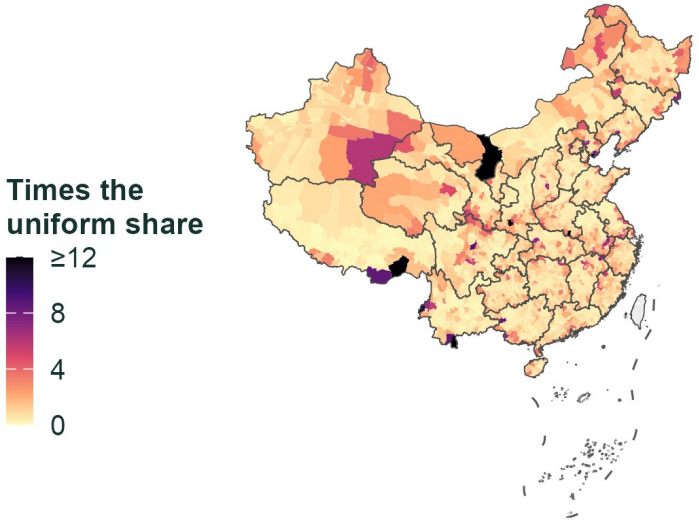
Checklists inside list 24% more species

More records ≠ better protection

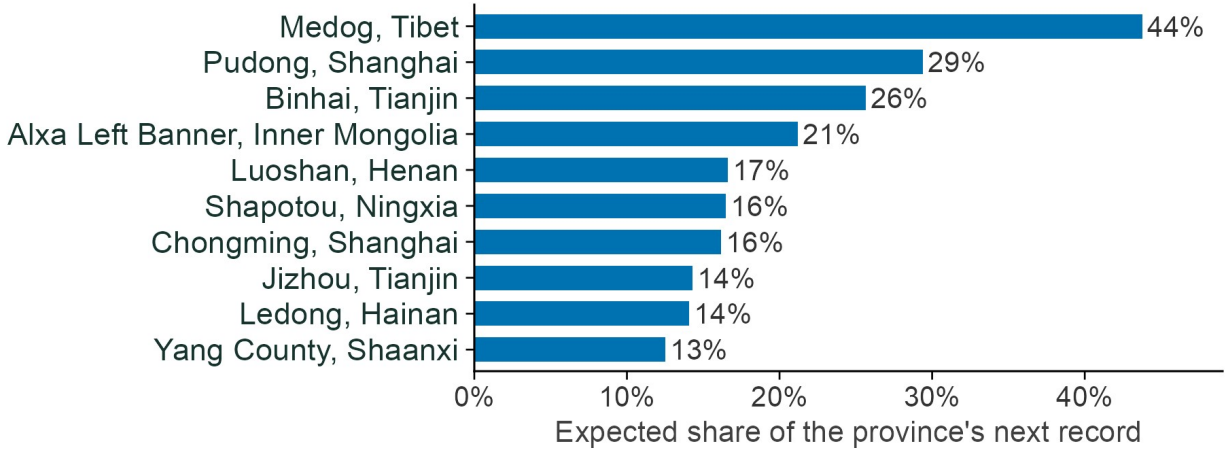
Record counts cannot measure how well a reserve works

Use 2 · Predictions can be pushed down to the county

a Relative chance of hosting the next provincial record



b Top 10 counties by expected share



Two steps

1 Will the province get a new record?

2 If so, which county? (conditional logit)

Out-of-sample skill, 2019–2024

Top-ranked county right 16.1% of the time

(random: about 1%)

True county in the top 10%: 45.4%

Leave one province out: 27 of 29 beat random

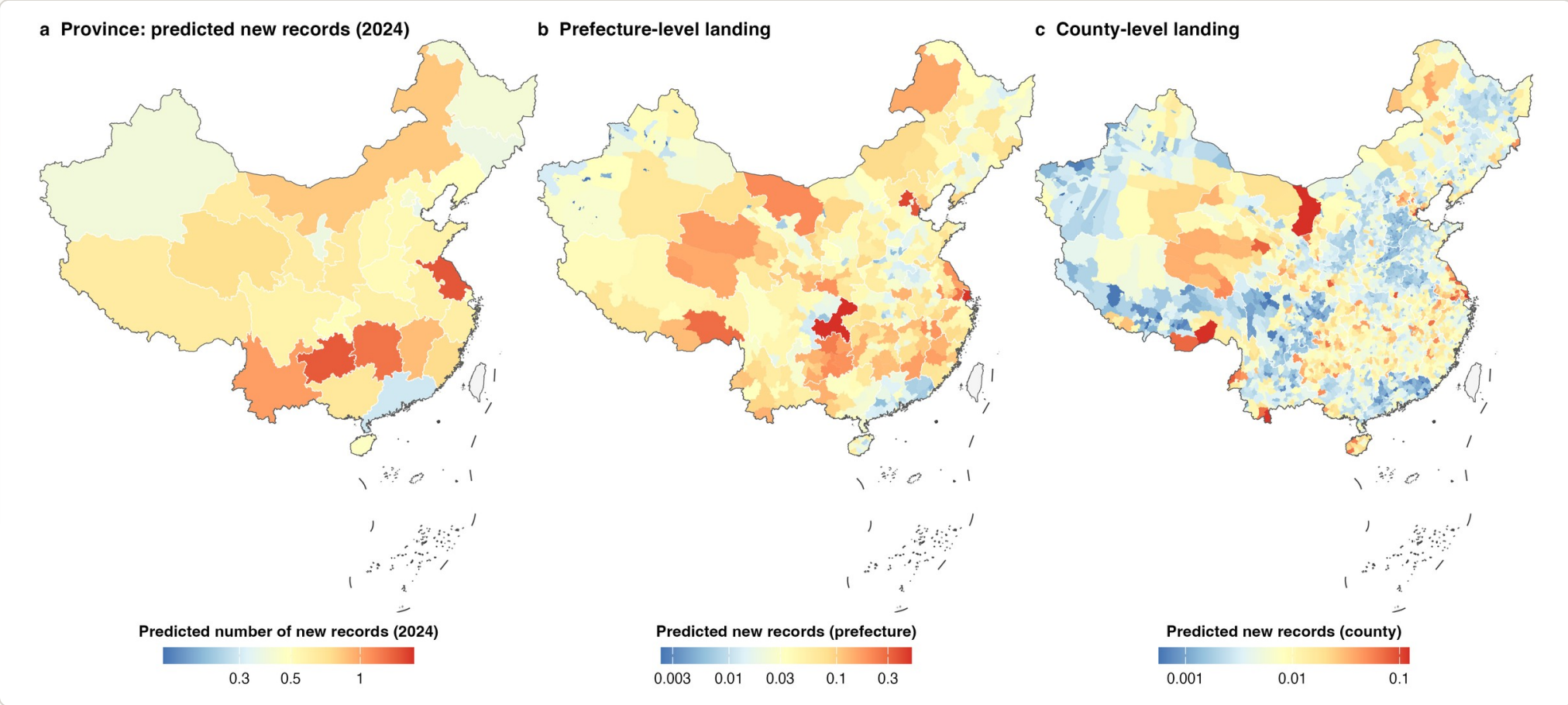
Birders will know these names

Medog · Pudong (Nanhui) · Chongming

Luoshan (Dongzhai) · Yang County

— the model was never told

Province, prefecture and county maps from one consistent model (2024)



One model, three scales

Expected records (province) × share (unit)

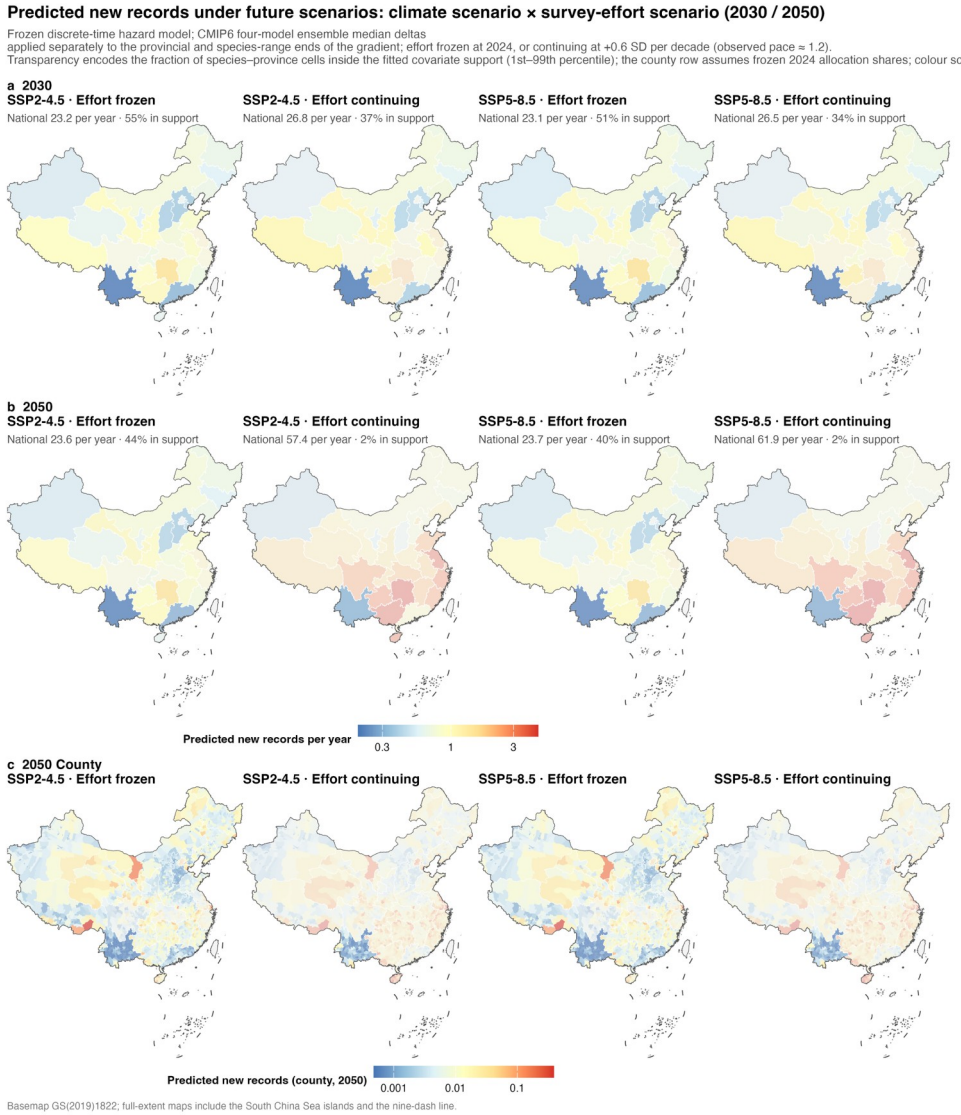
371 prefectures · 2,835 counties · log colour

Change the model family: maps barely move

Random forest / XGBoost: ρ = 0.62–0.90

Differences sit in the low-effort west

Futures: two climate × two effort scenarios (2030 and 2050)



Scenario grid

Climate: SSP2-4.5 or SSP5-8.5

Effort: frozen at 2024, or +0.6 SD a decade

Horizons: 2030 and 2050

What drives future record counts?

Switch climate: 23.2 → 23.1 per year

Switch effort: 23.6 → 57–62 per year

What we record will depend on how we look,

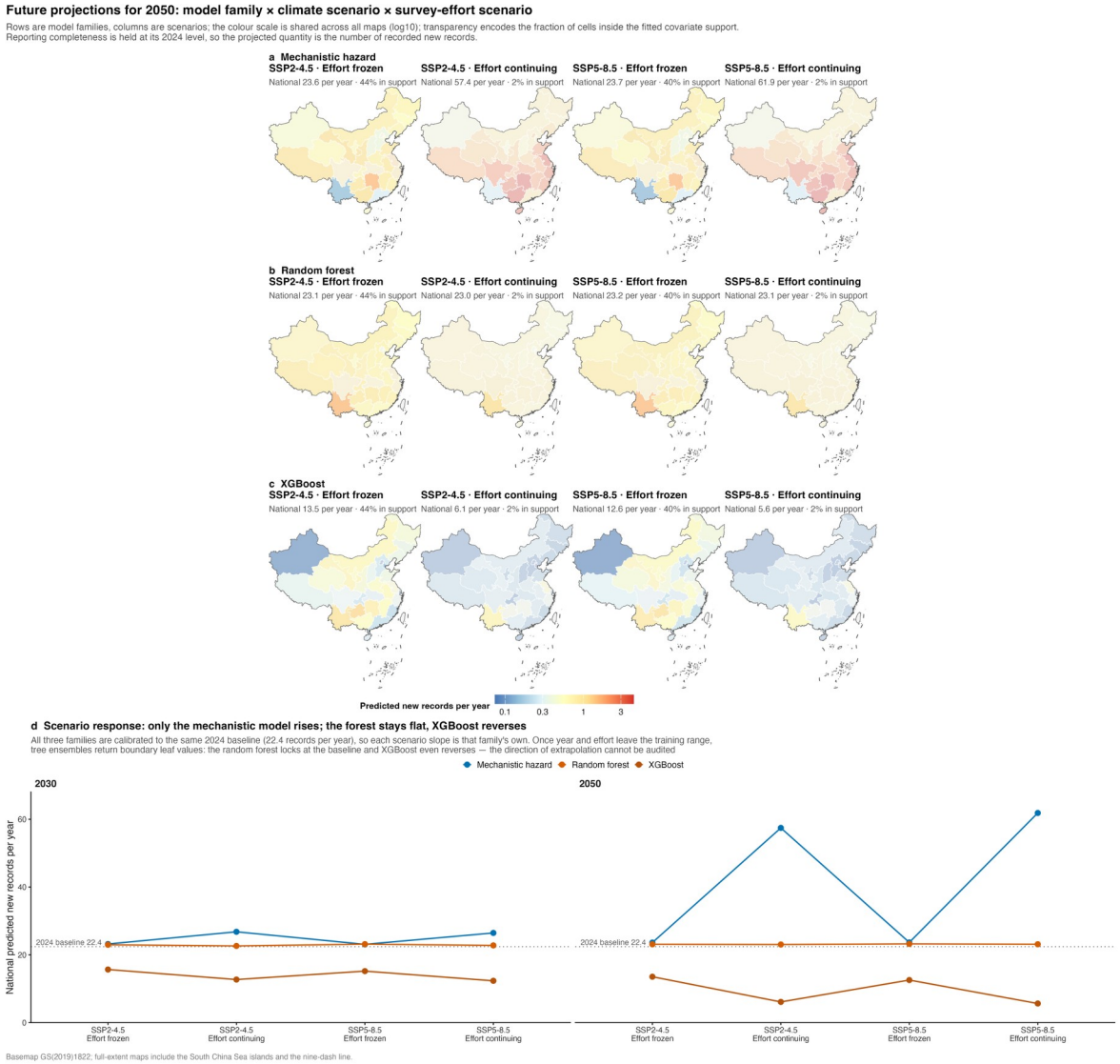
more than on how the climate changes

But read the transparency

Effort growing, 2050: only 2% in support

Most of the rise lies outside the data

Same futures, three model families: who responds, who flat-lines



Same scenarios, three model families

All calibrated to 22.4 records a year in 2024

After that, each family's slope is its own

2050, effort growing: they diverge

Hazard model: 23.6 → 57.4 per year

Random forest: 23.1 → 23.0 (flat)

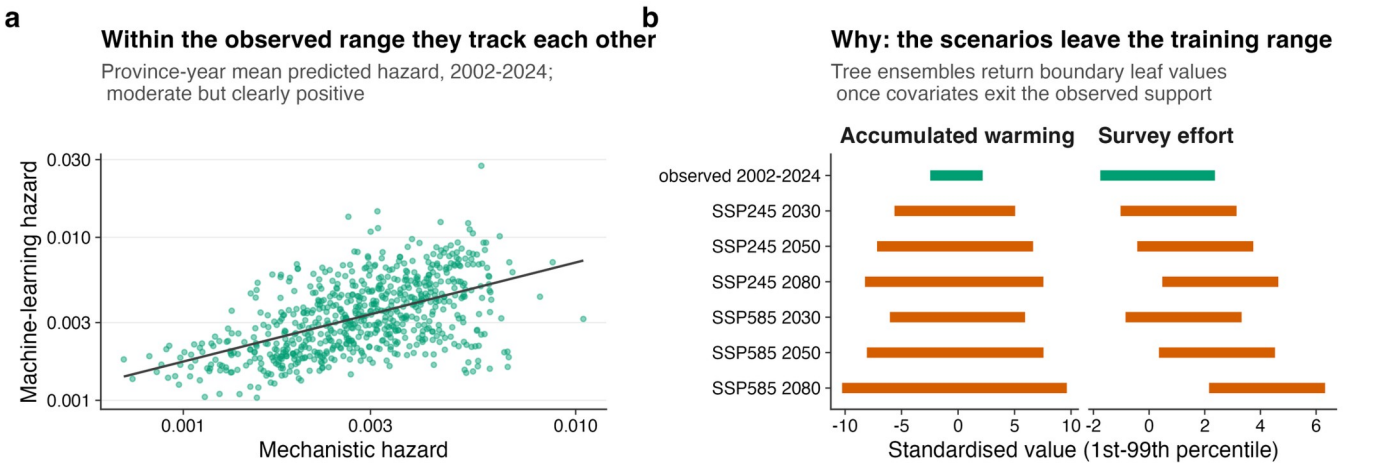
XGBoost: 13.5 → 6.1 (reversed)

Yet all agree on where (ρ = 0.62–0.90)

Not a contest between models

Outside the data, trees return edge values;
nobody can check the direction

Use 3 · Data support, not ambition, sets the forecast horizon



Data support shrinks fast (SSP2-4.5)

Species–province cells inside the fitted range:

2030: 47.8% → 2050: 10.4% → 2080: 2.1%

Masked HR 0.79–2.94; unmasked up to 78

Machine learning does not rescue it

Agree inside the range (ρ = 0.54),

diverge outside it

Random forest saturates at 6.7% by 2.9 SD

Report where a model breaks

2030: usable

End of century: illustration, not forecast

Four limits that travel with the conclusions

Events are published records, not detections

We explain how records are documented, not range expansion itself; single cases need occupancy models.

Coordinates are precise to about 1 km at best

Reserve-overlap tests therefore use 0–5 km buffers as a sensitivity check.

Reserve polygons cover about a third of listed reserves

1,028 mapped, mostly national, set up before 2012: no gap analysis, no 30×30 claims.

Direction and reserve tests are only as good as their nulls

Direction uses province-level geometric nulls; no null removes observer bias inside reserves.

Four take-home messages

1 Dating is a modelling decision

Publication dating inflates effort by 37%
and creates a phantom effect

2 Two processes, comparable and separable

Warming 1.36 · effort 1.40
increment over null ×1.18

3 Model the observation process

Province × year ×2.23 is the largest part;
effort saturates; warming shows no such trend

4 Keep applications inside the evidence

Records ≠ reserve success; counties work;
the future depends on effort

Birds or birders? Not either–or.

Both matter about equally — but the climate signal becomes interpretable only once we model how we look.

Next steps

Reptiles and amphibians · test the direction mechanism across taxa · occupancy models for single cases

Monitoring priority = climate exposure × (1 − current coverage), mapped to counties

Open data, open code — and thank you

Open data and code

CBNR (CC BY 4.0)

doi.org/10.5281/zenodo.20809949

Code: github.com/dingchenchen6/bird-newrecord-discovery-dated

Every number traces back to a results table

Other data

Ranges: BirdLife International

Climate: CRU TS 4.09 · WorldClim 2.1 · CMIP6

Effort: 2.86 million GBIF / eBird records

Reserves: 1,028 national nature-reserve maps

Base map: approval no. GS(2019)1822

Acknowledgements

Birdwatchers across China, and every author whose records made CBNR possible.

Thank you — questions welcome

chenchen.ding@pku.edu.cn

CBNR data

scan to download